

Assumption	Baseline	Sensitivity case 1	Sensitivity case 2	Source or basis	How it is used	Confidence level
Study period	50 years	50 years	50 years	Fixed project boundary	Defines the comparison period for both scenarios	High
Functional unit	One residential window opening with equivalent thermal, acoustic, safety and weather-protection performance	Same	Same	Project protocol	Keeps Scenario A and Scenario B comparable	High
Reference window size	1.23 x 1.48 m	1.20 x 1.40 m	1.23 x 1.48 m	VELFAC case documentation and project protocol	Defines the representative window opening for the scenario comparison	Medium
Frame life	40 years	30 years	50 years	Step 12 protocol assumption and manufacturer claim treated cautiously	Determines whether the long-life frame can plausibly be retained in Scenario B	Medium
Glazing life	25 years	20 years	35 years	Step 12 protocol assumption and literature-based service-life logic	Determines whether glazing replacement may be needed before the full frame reaches end of life	Medium
Hardware replacement interval	15 years	10 years	20 years	Step 12 protocol assumption and interview evidence that mechanisms and handles wear first	Determines likely timing of hardware, handle or fitting replacement in Scenario B	Medium
Seal or gasket replacement interval	20 years	10 years	25 years	Assumption based on short-life component logic and interview evidence	Tests whether smaller components may need replacement before full-window replacement	Low to medium
Full replacement interval in Scenario A	25 years	20 years	30 years	Scenario assumption based on comparison logic and interview reference to older systems around 25 years	Determines when a complete replacement window is added in Scenario A	Medium
Repair events in Scenario B	2 events	1 event	3 events	Step 12 protocol assumption	Represents number of maintenance or component-replacement interventions over 50 years	Medium
Transport distance	100 km	50 km	300 km	Step 12 protocol assumption	Tests whether repair or replacement benefit may be affected by transport distance	Low to medium
Service visit grouping	Repairs grouped where possible	All repairs grouped	Repairs separate	Scenario assumption	Tests whether multiple repair visits may increase transport and labour burden	Low
End-of-life route	Recycling where available, otherwise mixed recovery	More favourable recycling	Less favourable recycling or uncertain recovery	Case documentation and literature coding on recycling uncertainty	Affects whether full replacement or component replacement shifts burden to end-of-life stages	Low
Spare-part availability	Available for normal service period but uncertain long-term	Available beyond 25 years	Limited after 25 years	Supplier interview and case documentation uncertainty	Determines whether Scenario B remains practical over time	Medium

Component access	Partly accessible	Good access	Poor access	Case documentation and interviews	Determines whether repairable components are actually reachable in practice	Medium
Direct reuse potential	Low	Medium if dimensions match	Very low if dimensions are project-specific	Supplier and architect interviews	Explains why direct reuse is excluded from main scenario comparison	Medium
Environmental impact values	Not quantified	Not quantified	Not quantified	No reliable product-specific values for repair events and use-phase assumptions	Prevents false precision and keeps the study exploratory	High